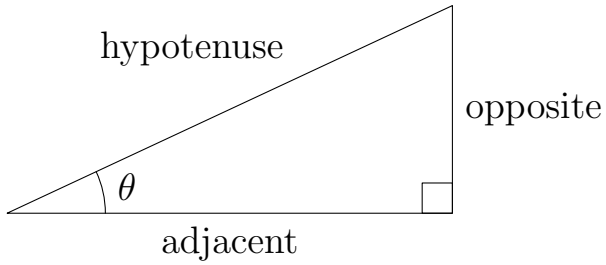
5.4: Right Angle Trigonometry

Definition.
Given a right triangle with an acute angle θ



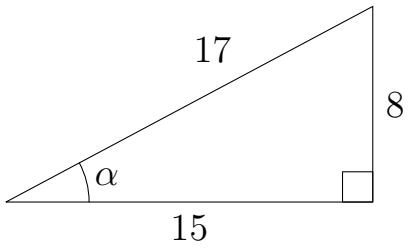
we have

$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$$

Example. Given the following triangle, find $\sin(\alpha)$, $\cos(\alpha)$, and $\tan(\alpha)$:



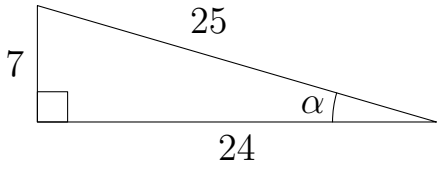
SOH $\quad \sin\alpha = \frac{8}{17}$

CAH $\quad \cos\alpha = \frac{15}{17}$

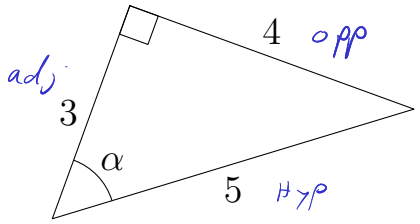
TOA $\quad \tan\alpha = \frac{8}{15}$

Example. Given the following triangle, find $\sin(\alpha)$:

SOH $\quad \sin\alpha = \frac{7}{25}$



Example. Given the following triangle, evaluate:

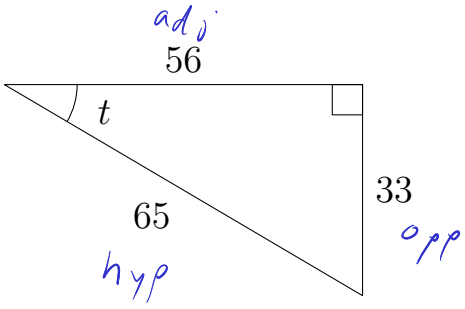


SOH $\quad \sin(\alpha) = \frac{4}{5}$ $\qquad \csc(\alpha) = \frac{5}{4}$

CAH $\quad \cos(\alpha) = \frac{3}{5}$ $\qquad \sec(\alpha) = \frac{5}{3}$

TOA $\quad \tan(\alpha) = \frac{4}{3}$ $\qquad \cot(\alpha) = \frac{3}{4}$

Example. Given the following triangle, evaluate:

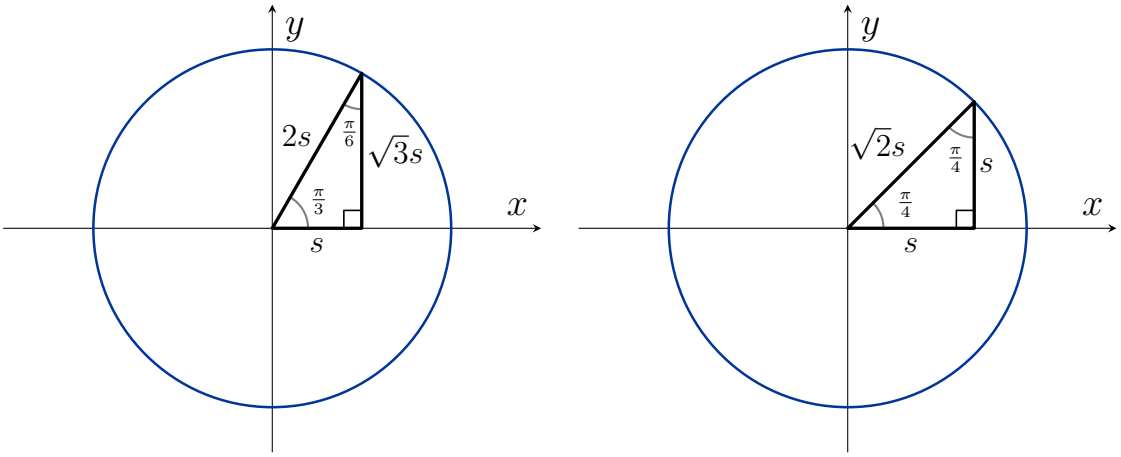


SOH $\quad \sin(\alpha) = \frac{33}{65}$ $\qquad \csc(\alpha) = \frac{65}{33}$

CAH $\quad \cos(\alpha) = \frac{56}{65}$ $\qquad \sec(\alpha) = \frac{65}{56}$

TOA $\quad \tan(\alpha) = \frac{33}{56}$ $\qquad \cot(\alpha) = \frac{56}{33}$

Side lengths of special angles:



Example. Find the exact value of the trigonometric functions of $\theta = \dfrac{\pi}{3}$, using side lengths.

$\sin(\theta) = \dfrac{\sqrt{3}s}{2s} = \boxed{\dfrac{\sqrt{3}}{2}}$

$\cos(\theta) = \dfrac{s}{2s} = \boxed{\dfrac{1}{2}}$

$\tan(\theta) = \dfrac{\sqrt{3}s}{s} = \boxed{\sqrt{3}}$

$\csc(\theta) = \dfrac{2s}{\sqrt{3}s} = \dfrac{2}{\sqrt{3}} = \boxed{\dfrac{2\sqrt{3}}{3}}$

$\sec(\theta) = \dfrac{2s}{s} = \boxed{2}$

$\cot(\theta) = \dfrac{s}{\sqrt{3}s} = \dfrac{1}{\sqrt{3}} = \boxed{\dfrac{\sqrt{3}}{3}}$

Example. Find the exact value of the trigonometric functions of $\vartheta = \dfrac{\pi}{4}$, using side lengths.

$\sin(\vartheta) = \dfrac{s}{\sqrt{2}s} = \dfrac{1}{\sqrt{2}} = \dfrac{\sqrt{2}}{2}$

$\cos(\vartheta) = \dfrac{s}{\sqrt{2}s} = \dfrac{1}{\sqrt{2}} = \dfrac{\sqrt{2}}{2}$

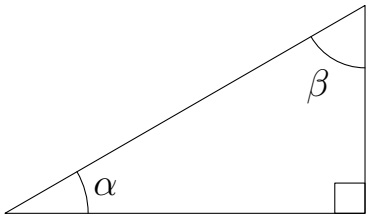
$\tan(\vartheta) = \dfrac{s}{s} = 1$

$\csc(\vartheta) = \dfrac{\sqrt{2}s}{s} = \sqrt{2}$

$\sec(\vartheta) = \dfrac{\sqrt{2}s}{s} = \sqrt{2}$

$\cot(\vartheta) = \dfrac{s}{s} = 1$

Equal cofunctions of complements:



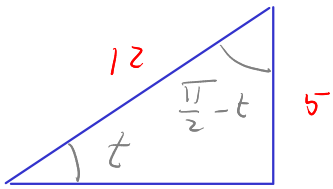
$$\sin(\alpha) = \cos(\beta)$$
$$\sin(\beta) = \cos(\alpha)$$

Definition. (Cofunction Identities)

$$\cos(\theta) = \sin\left(\frac{\pi}{2} - \theta\right) \qquad \sin(\theta) = \cos\left(\frac{\pi}{2} - \theta\right)$$
$$\sec(\theta) = \csc\left(\frac{\pi}{2} - \theta\right) \qquad \csc(\theta) = \sec\left(\frac{\pi}{2} - \theta\right)$$
$$\cot(\theta) = \tan\left(\frac{\pi}{2} - \theta\right) \qquad \tan(\theta) = \cot\left(\frac{\pi}{2} - \theta\right)$$

Example. If $\sin(t) = \frac{5}{12}$, find $\cos\left(\frac{\pi}{2} - t\right)$.

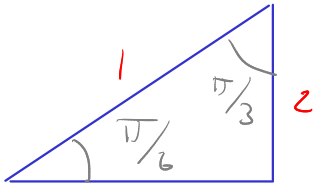
$\cos\left(\frac{\pi}{2} - t\right) = \boxed{\frac{5}{12}}$



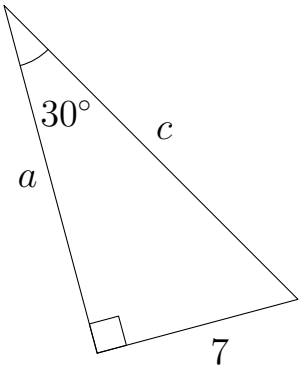
Example. If $\csc\left(\frac{\pi}{6}\right) = 2$, find $\sec\left(\frac{\pi}{3}\right)$.

$\frac{\text{Hyp}}{\text{Opp}}$ $\frac{\text{Hyp}}{\text{Adj}}$

$\sec\left(\frac{\pi}{3}\right) = \frac{2}{1} = \boxed{2}$



Example. Find the unknown sides of the following triangle:



$\sin(30°) = \frac{7}{c}$ $\qquad$ $\tan(30°) = \frac{7}{a}$

$\frac{1}{2} = \frac{7}{c}$ $\qquad$ $\frac{\sqrt{3}}{3} = \frac{7}{a}$

$\Rightarrow \boxed{c = 14}$ $\qquad$ $\Rightarrow \boxed{a = 7\sqrt{3} \approx 12.1244}$

Example. To find the height of a tree, a person walks to a point 30 feet from the base of the tree. She measures an angle of 57° between a line of sight to the top of the tree and the ground. Find the height of the tree.

$\tan(57°) = \frac{h}{30} \Rightarrow \boxed{h = 30\tan(57°) \approx 46.1959 \text{ ft}}$

